TTK4115 Helicopter Lab Prep

Part II - Multivariable Control

Problem 1 - State space formulation

Bring the system

$$\ddot{p} = K_1 V_d$$
$$\ddot{e} = K_2 \tilde{V}_s$$

On the form

$$\dot{x} = Ax + Bu$$

with

$$x = \begin{pmatrix} p \\ \dot{p} \\ \dot{e} \end{pmatrix} \quad \wedge \quad u = \begin{pmatrix} \tilde{V}_s & V_d \end{pmatrix}$$

which becomes

$$\begin{pmatrix} \dot{p} \\ \ddot{p} \\ \ddot{e} \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} p \\ \dot{p} \\ \dot{e} \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & K_1 \\ K_2 & 0 \end{pmatrix} \begin{pmatrix} \tilde{V}_s & V_d \end{pmatrix}$$

with the matrices

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \wedge \quad B = \begin{pmatrix} 0 & 0 \\ 0 & K_1 \\ K_2 & 0 \end{pmatrix}$$

Problem 2 - Controllability

Examining the controllability matrix $\mathcal{C}$

$$\begin{aligned} \mathcal{C} &= (B \,|\, AB \,|\, A^2B) \\ &= \left(\begin{array}{cc|cc|cc} 0 & 0 & 0 & K_1 & 0 & 0 \\ 0 & K_1 & 0 & 0 & 0 & 0 \\ K_2 & 0 & 0 & 0 & 0 & 0 \end{array}\right) \end{aligned}$$

and since $\text{rank}(\mathcal{C}) = 3$, the system is controllable.

Checking in `MATLAB` using `ctrb(A,B)` gives the same matrix, with rank 3.

Problem 3 - Feedback and feedforward

Track the reference $r = \begin{pmatrix} p_c \\ \dot{e}_c \end{pmatrix}$

with the control law

$$u = Fr - Kx$$

where

$$K = \begin{pmatrix} k_{11} & k_{12} & k_{13} \\ k_{21} & k_{22} & k_{23} \end{pmatrix} \quad \wedge \quad F = \begin{pmatrix} F_{11} & F_{12} \\ F_{21} & F_{22} \end{pmatrix}$$

Inserting this into the system gives

$$
\begin{aligned}
\begin{pmatrix} \dot{p} \\ \ddot{p} \\ \ddot{e} \end{pmatrix} &= \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} p \\ \dot{p} \\ \dot{e} \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & K_1 \\ K_2 & 0 \end{pmatrix} \left(\begin{pmatrix} F_{11} & F_{12} \\ F_{21} & F_{22} \end{pmatrix} \begin{pmatrix} p_c \\ \dot{e}_c \end{pmatrix} - \begin{pmatrix} k_{11} & k_{12} & k_{13} \\ k_{21} & k_{22} & k_{23} \end{pmatrix} \begin{pmatrix} p \\ \dot{p} \\ \dot{e} \end{pmatrix} \right) \\
&= \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} p \\ \dot{p} \\ \dot{e} \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & K_1 \\ K_2 & 0 \end{pmatrix} \left(\begin{pmatrix} p_c F_{11} + p_c F_{12} \\ \dot{e}_c F_{21} + \dot{e}_c F_{22} \end{pmatrix} - \begin{pmatrix} k_{11} p + k_{12} \dot{p} + k_{13} \dot{e} \\ k_{21} p + k_{22} \dot{p} + k_{23} \dot{e} \end{pmatrix} \right) \\
&= \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} p \\ \dot{p} \\ \dot{e} \end{pmatrix} + \begin{pmatrix} 0 \\ K_1(\dot{e}_c F_{21} + \dot{e}_c F_{22}) - K_1(k_{21} p + k_{22} \dot{p} + k_{23} \dot{e}) \\ K_2(p_c F_{11} + p_c F_{12}) - K_2(k_{11} p + k_{12} \dot{p} + k_{13} \dot{e}) \end{pmatrix} \\
&= \begin{pmatrix} 0 & 1 & 0 \\ -K_1 k_{21} & -K_1 k_{22} & -K_1 k_{23} \\ -K_2 k_{11} & -K_2 k_{12} & -K_2 k_{13} \end{pmatrix} \begin{pmatrix} p \\ \dot{p} \\ \dot{e} \end{pmatrix} + \begin{pmatrix} 0 \\ K_1(\dot{e}_c F_{21} + \dot{e}_c F_{22}) \\ K_2(p_c F_{11} + p_c F_{12}) \end{pmatrix}
\end{aligned}
$$

as $t \to \infty$, this becomes

$$
\begin{aligned}
0 &= \begin{pmatrix} 0 & 1 & 0 \\ -K_1 k_{21} & -K_1 k_{22} & -K_1 k_{23} \\ -K_2 k_{11} & -K_2 k_{12} & -K_2 k_{13} \end{pmatrix} \begin{pmatrix} p \\ 0 \\ \dot{e} \end{pmatrix} + \begin{pmatrix} 0 \\ K_1 \dot{e}_c (F_{21} + F_{22}) \\ K_2 p_c (F_{11} + F_{12}) \end{pmatrix} \\
&= \begin{pmatrix} -K_1 k_{21} - K_1 k_{23} \\ -K_2 k_{11} - K_2 k_{13} \end{pmatrix} \begin{pmatrix} p \\ \dot{e} \end{pmatrix} + \begin{pmatrix} 0 & K_1(F_{21} + F_{22}) \\ K_2(F_{11} + F_{12}) & 0 \end{pmatrix} \begin{pmatrix} p_c \\ \dot{e}_c \end{pmatrix}
\end{aligned}
$$

for $p \to p_c$ and $\dot{e} \to \dot{e}_c$:

$$
F_{21} + F_{22} = k_{23} \quad \wedge \quad F_{11} + F_{12} = k_{11}
$$

so F becomes, for instance:

$$
F = \begin{pmatrix} k_{11} & 0 \\ 0 & k_{23} \end{pmatrix}
$$

Problem 4

Larger "Q" matrix will lead to more aggressive control, and larger "R" matrix will lead to more smooth control.

Problem 5

The new A, B, and G matrices are simply:

$$
A = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 \end{pmatrix} \quad \wedge \quad B = \begin{pmatrix} 0 & 0 \\ 0 & K_1 \\ K_2 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix} \quad \wedge \quad G = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}
$$

with the added states γ and ζ the Q matrix will also penalize the error in the states, hence helping drive them to 0. The G matrix will help drive the states to the reference. The introduction of integral effect makes it less important to choose F appropriately